# Sean J. Moran, PhD

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## AI/ML Managing Director

Enterprise AI Transformation | LLM & Multi-Agent Systems | AI Governance | Research-to-Production

Public CV. Full CV available on request.

## Profile

AI/ML leader specialising in enterprise AI transformation — LLM/RAG/multi-agent automation (including Model Context Protocol), evaluation-driven deployment, and strong AI governance — built on deep foundations in NLP, computer vision, and information retrieval. Former Executive Director of AI/ML at a global investment bank; now Head of AI, UK at a global investment and technology group, leading applied research and production AI in regulated environments while remaining up to **50% hands-on**. PhD in Artificial Intelligence from the University of Edinburgh.

## Selected Experience

**Global Investment & Technology Group** — **London, UK**
***Head of AI, UK / Managing Director, AI/ML*** — *2025 – Present*

- Lead the UK AI function covering applied AI, data science, generative AI, enterprise automation, and regulated-domain AI delivery.
- Built and manage a UK-based AI/data science team while remaining hands-on across solution design, prototyping, architecture, and technical delivery.
- Partner with senior stakeholders across financial services, insurance, healthcare, technology, and other regulated enterprise domains.
- Lead AI discovery, technical workshops, solution architecture, rapid prototype development, and productionisation planning; technical pre-sales work has helped convert opportunities into funded engagements and follow-on programmes.
- Delivered initiatives across multi-agent AI, document intelligence, fraud analytics, enterprise knowledge assistants, predictive modelling, and AI-enabled operational automation.
- **Core AI expertise:** LLM fine-tuning & post-training, RAG, multi-agent systems, LLM evaluation frameworks; frontier-model APIs (GPT-4/5, Claude, Gemini, Llama); classical ML where appropriate.
- **Tooling & platforms:** Python, PyTorch, Scikit-learn, LangChain, Model Context Protocol (MCP), AWS.

**Global Investment Bank** — **London, UK**
***Executive Director, AI/ML*** — *2020 – 2025*

- Founded and led a large applied-AI organisation of data scientists, engineers, and product specialists delivering production-grade AI for technology and operations.
- Member of senior technology leadership forums, responsible for technical direction, delivery governance, prioritisation, team development, and investment planning.
- Delivered enterprise AI systems for incident analysis, enterprise search, technology risk, software intelligence, vulnerability triage, infrastructure analytics, and operational automation.
- Led generative-AI and graph-augmented retrieval initiatives used by large internal user populations.
- Recognised for innovation, applied-AI research, patent output, open-source contributions, and production delivery in highly regulated environments.

**Global Technology Company** — **London, UK**
***Senior Research Scientist*** — *2018 – 2020*

- Led applied research in computer vision and computational photography, including image enhancement, low-light imaging, denoising, and neural image-processing pipelines.
- Developed deep-learning architectures for image and video enhancement under adverse conditions.
- Published research at major computer-vision venues and filed patents in neural image processing.

**Biotechnology / Gene Therapy Company** — **Edinburgh, UK**
***Machine Learning Engineer*** — *2016 – 2018*

- First machine-learning hire, building AI and bioinformatics capabilities for regulatory genomics and precision medicine.
- Developed ML pipelines for modelling DNA regulatory sequences and predicting cell-type-specific gene expression.
- Built end-to-end data-processing, modelling, and visualisation tools for scientific users.

**University of Glasgow** — **Glasgow, UK**
***Data Science Consultant (Contract)*** — *2015 – 2016*

- Consultant in the Terrier Information Retrieval group on the ESRC-funded *Integrated Multimedia City Data* Urban Big Data Centre project.
- Integrated and correlated social media, GPS, weather, and transport datasets to model urban mobility; built a transport-mode classifier and improved Twitter first-story detection using Word2Vec semantic similarity.
- Deliberately short fixed-term contract following PhD completion; declined an offered extension to return to industry.

**University of Edinburgh** — **Edinburgh, UK**
***PhD Researcher, Artificial Intelligence*** — *2011 – 2015*

- Doctoral research in large-scale information retrieval and computer vision: learning-to-hash, approximate nearest-neighbour search, and probabilistic relevance modelling for image annotation.
- Built a low-latency streaming event-detection system processing 4M tweets/day; released open-source toolkits still cited and reused by the research community; Best Student Paper, ICMR 2014.

**University of Edinburgh** **Edinburgh, UK**
*Data Science Consultant (Contract)* *2010*

- Consultant in the Information Retrieval Lab, partnering with a national broadcaster's R&D team on cross-modal retrieval and video understanding.
- Designed and built an end-to-end video-to-Wikipedia alignment pipeline — an early cross-modal retrieval system combining shot-boundary and on-screen-text detection, face detection, and speech segmentation/transcription.
- Short fixed-term contract immediately preceding the PhD.

**Tier-1 Investment Bank** **London, UK**
*Software Engineer, Commodities Trading* *2005 – 2008*

- Lead engineer building mission-critical .NET and Java applications for the Commodities Trading franchise; promoted to Associate within one year; led a development team while working directly with traders, analysts, and front-office management.
- Designed and delivered a 3-tier European power-scheduling platform for day-ahead and intra-day scheduling across multiple European power markets; deployed bank-wide under strict latency, reliability, and regulatory constraints.
- Built an auto-trading agent that monitored market spreads and automatically submitted bids in the European gas market — an early application of rule-based, event-driven automation in finance.

## Education

- **PhD in Artificial Intelligence** – University of Edinburgh. Research focus: learning to hash, large-scale image retrieval, representation learning, information retrieval.
- **MSc in Artificial Intelligence, Distinction** – University of Edinburgh. Awarded School of Informatics prize for outstanding achievement.
- **MA in Computer Science & Management Studies** – University of Cambridge.

## Technical Focus

- **AI / ML:** Generative AI, LLM fine-tuning & post-training, RAG, multi-agent systems, LLM evaluation frameworks, information retrieval, search, recommender systems, computer vision, NLP, deep learning, classical ML, responsible AI & AI governance.
- **Engineering:** Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, LlamaIndex, Model Context Protocol (MCP), MLflow, Databricks, AWS, Azure, OpenCV, Java, JavaScript, SQL.
- **Leadership:** AI strategy, team building, technical governance, executive stakeholder management, applied research, productisation, pre-sales support, regulated-enterprise delivery.

## Research, Patents & Open Source

- 40+ peer-reviewed publications and 1,200+ academic citations; full publication list via **Google Scholar**.
- 50+ patent filings (25+ granted) across applied AI, NLP, code intelligence, retrieval, federated learning, and enterprise automation; complete patent portfolio at **sjmoran.github.io/#patents**.
- Creator and maintainer of open-source AI projects in image enhancement, learning-to-hash, LLM research indexing, and applied AI tooling.
- Public work available via Google Scholar, GitHub, and personal portfolio.

## Selected Recognition

- Enterprise innovation awards for applied AI and generative-AI software-engineering systems.
- Best Student Paper, ICMR 2014, for probabilistic relevance modelling for image annotation.
- EPSRC Doctoral Scholarship, PhD in Artificial Intelligence, University of Edinburgh.
- School of Informatics Prize for Outstanding Achievement, University of Edinburgh.
- St Catharine's College Thomas Hobbes Scholarship, University of Cambridge.